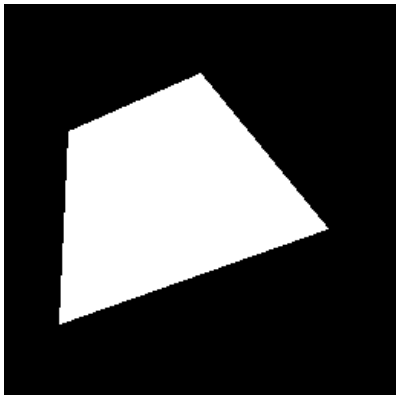
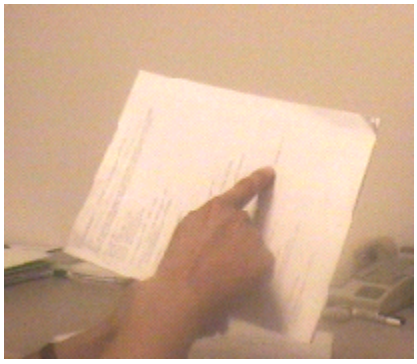
Nolan Knight
MSAI495
MP#5
April 30, 2026

In this problem, we implemented Hough Transforms using Carry Edge Detection to extract the edges from the test images and identify lines that are part of the same image. The following sample images were used for the training and testing:



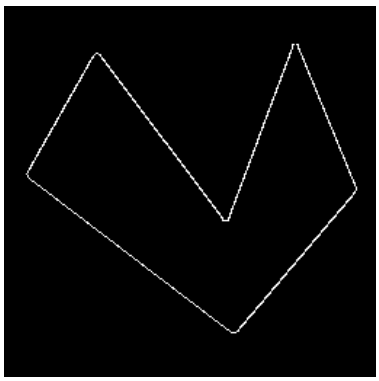
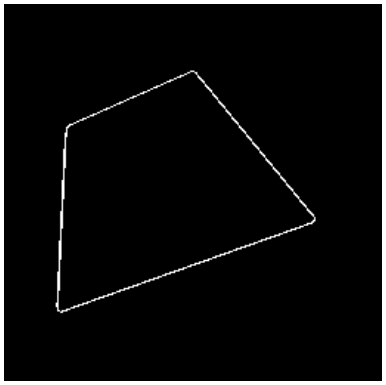


This problem was broken up into 4 Steps:

1) Edge Detection: (Carry Edge Detection MP#4)
2) Polar Coordinate Parameter Space (rho and theta)
    a) Rho: Distance from Origin
    b) Theta: Normal to the line
3) Voting:
    a) For each coordinate in the parameter space that is an Edge, vote on the line that could be formed through that point
4) Max Peaks in Voting:
    a) Determine which points in the parameter space are most likely to be a line in the image.
5) Draw the Lines:
    a) Convert the parameter space local maximums back to the image space and plot the identified lines.

For each step the following outputs were obtained:

Step 1: Edge Detection:

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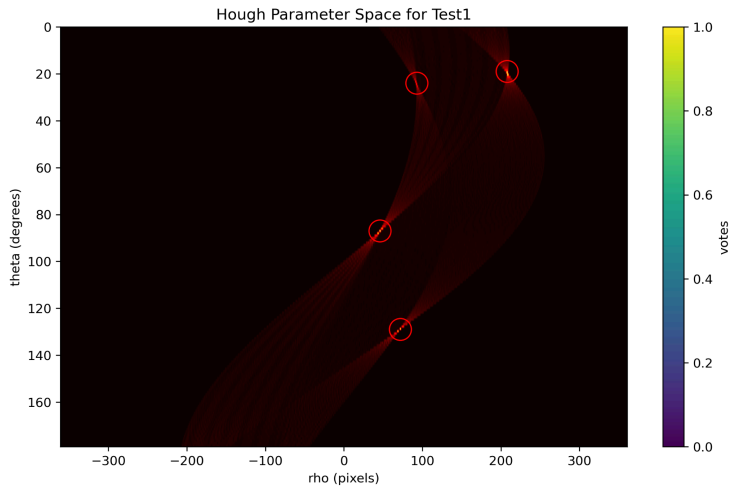
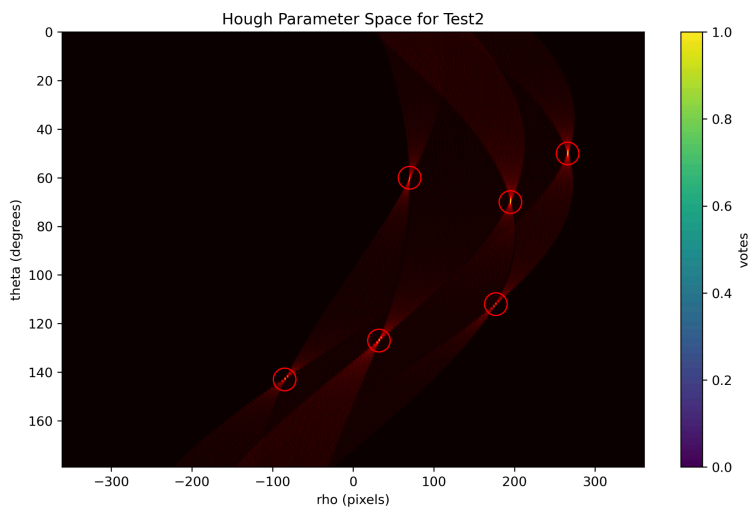
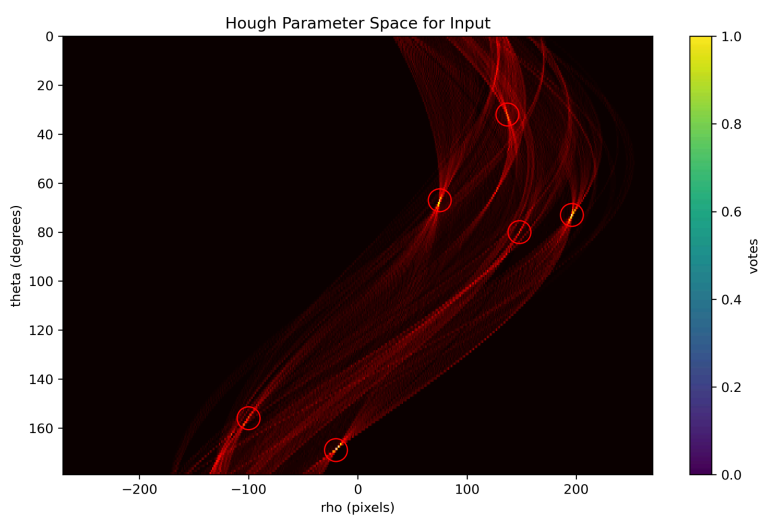
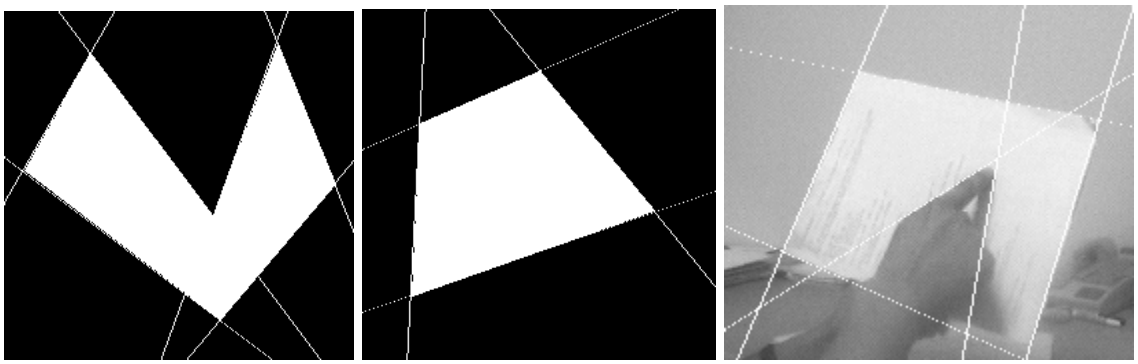
Step 2-4: Parameter Space and Voting (Red Circle)



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Step 5: Draw the Lines on Image: